Technical University of Denmark

02452 - PROJECT 2

Case: Gallstone Disease

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Disclaimer: AI tools were used in this project as assistance for text formulation and as guidance in developing and refining Python code.

CONTRIBUTION TABLE

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1 Regression, Part a

1.1 Linear Model

This project continues the analysis of the Gallstone dataset introduced in project 1. Here, we extend the work by applying both regression and classification techniques to further investigate how biological factors are related to gallstone disease and its associated biological markers.

1.1.1 Regression Setup

In this part, a linear regression analysis is performed to explore how different biological factors influence the concentration of C-reactive protein (CRP). We have chosen CRP as our target variable as elevated levels are known to reflect inflammatory activity, which is associated with the development of gallstones. The purpose of this analysis is therefore to identify which biological factors most strongly affect CRP levels. The predictors (X) used in the regression model are listed in Table 1. These attributes were chosen for their relevance to inflammation and metabolic health, both linked to gallstone disease.

No.	Attribute	Description
1	Gender	Biological sex (0 = female, 1 = male)
2	Age	Age of the participant (years)
3	Height	Body height (cm)
4	Weight	Body weight (kg)
5	Vitamin D	Vitamin D concentration (nmol/L)
6	TBFR	Total Body Fat Ratio (%)
y	CRP	C-reactive protein concentration (mg/L)

Table 1: *Attributes included in the linear regression model predicting CRP concentration.*

Before modeling, the dataset was preprocessed. Gender was already binary encoded (0/1), so no additional encoding was needed. Furthermore, all variables were standardized so that each column in the data matrix X had a mean of 0 and standard deviation of 1, following the transformation:

$$x' = \frac{x - \mu}{\sigma}$$

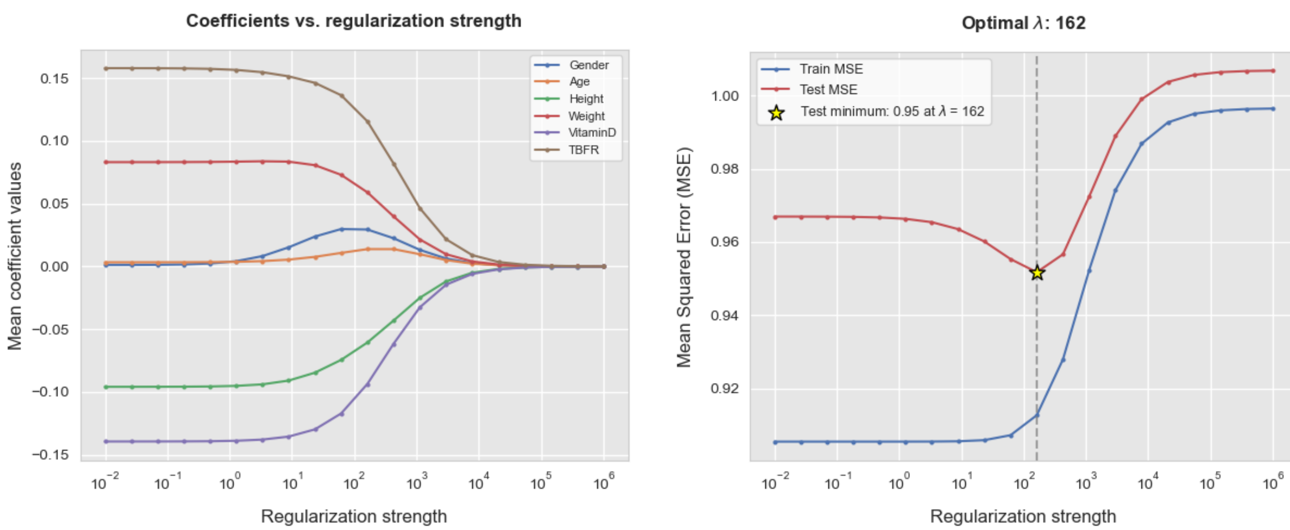
These preprocessing steps had already been completed in project 1 and were therefore not repeated here. If categorical features with more than two levels had been included, one-of-K encoding would have been applied.

1.1.2 Regularization and Cross-Validation

To improve model generalization and prevent overfitting, a regularization parameter (λ) was introduced. Regularization balances bias and variance, enabling the model to capture true patterns without fitting noise.

Because biological features are often correlated, the regression model can produce unstable parameter estimates. L2 Regularization addresses this by penalizing large coefficients ($\mathbf{w}$), which reduces variance and improves generalization. As illustrated in Figure 1a, increasing λ constrains the coefficients and lowers variance, while smaller values allow the model to fit the training data more closely, resulting in lower bias but higher variance.

The optimal λ was determined using 10-fold cross-validation. For each value of λ, the model was trained on nine folds and evaluated on the remaining fold. This process was repeated so that each fold served as validation once. The average Mean Squared Error (MSE) across the 10 validation folds was used as an estimate of the generalization error. To ensure that we captured the full trade-off between underfitting and overfitting, the regularization parameter was selected from a logarithmic range $\lambda \in [10^{-2}, 10^{6}]$. Using this range allowed us to observe the expected U-shaped validation error curve and identify the optimal degree of regularization. As illustrated in Figure 1b, the test error initially decreases with increasing λ, reaching a minimum at $\lambda = 162$, before rising again for larger values due to increased bias.



(a) (b)

Figure 1: *(a) Effect of the regularization parameter λ and (b) estimated training and test errors as functions of λ, obtained using 10-fold cross-validation across a logarithmic range of $\lambda \in [10^{-2}, 10^{6}]$.*

1.1.3 Interpretation of the Linear Model

The output $\hat{y}$ of the linear regression model with the lowest generalization error (i.e., the optimal regularization parameter λ) is computed as a weighted sum of the standardized input features plus an intercept.

The model can be written as:

$$\hat{y} = w_0 + \sum_i w_i x_i = \tilde{\mathbf{x}}^\top \mathbf{w}^*$$

where x_i are the standardized feature values, w_i are the corresponding coefficients (weights), and w_0 is the intercept.

The model parameters were estimated using the closed-form solution for ridge regression.

$$\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

Here, the λ value selected through cross-validation determines how strongly the coefficients are regularized.

Each coefficient w_i represents how much the predicted output $\hat{y}$ changes for a one standard deviation increase in the corresponding feature x_i. A positive weight increases the predicted CRP level, while a negative weight decreases it. For the optimal model $\lambda = 162$, the estimated weights are shown in Table 2.

Feature x_i	**Weight** w_i
Vitamin D	-0.1129
TBFR	0.3996
Weight	0.1375
Height	-0.2356
Gender	0.0963
Age	-0.1239

Table 2: *Estimated feature weights for the optimal ridge regression model with regularization parameter* $\lambda = 162$.

The estimated weights show that TBFR has the strongest positive association with CRP, indicating that higher body fat is linked to increased inflammation. In contrast, Vitamin D and Height have negative effects, while Weight, Age, and Gender show smaller positive contributions.

These findings align with physiological expectations: CRP is typically elevated with higher adiposity [1], whereas Vitamin D has anti-inflammatory properties [2]. Thus, the model demonstrates both good predictive performance and biological interpretability.

Overall, the ridge regression model achieved a low generalization error and produced interpretable coefficients that align with established biological relationships. The applied regularization enhanced model stability by mitigating the effects of correlated features, thereby increasing confidence in the obtained results.

2 Regression, Part b

In this section, we extend the previous ridge regression analysis by comparing three different regression models using a two-level cross-validation framework. Specifically, we compare three models: a ridge regression model, an artificial neural network (ANN), and a simple baseline that predicts the mean of the training data. The comparison aims to evaluate which model generalizes best to unseen data and whether any performance differences are statistically significant.

2.1 Two-Level Cross-Validation

A two-level cross-validation was implemented to ensure an unbiased estimate of the generalization error while simultaneously tuning the model hyperparameters. This approach enables fair model comparison by separating hyperparameter optimization (inner loop) from final evaluation (outer loop). Both the outer and inner loops used $K_1 = K_2 = 10$ folds. In each outer loop, the dataset was divided into a training partition ($\mathcal{D}_i^{par}$) and a held-out test partition ($\mathcal{D}_i^{test}$). Within each outer fold, the inner loop was used to determine the optimal hyperparameters: the regularization strength λ for Ridge regression and the number of hidden units h for the ANN model. The baseline model simply predicted the mean of y_{train} from the corresponding training fold.

For the Ridge regression model, the hyperparameter search was conducted over the same logarithmic range as in Part 1, $\lambda \in [10^{-2}, 10^6]$ (13 logarithmically spaced values), to ensure consistency and allow direct comparison between the two analyses. For the ANN, the model complexity was controlled by varying the number of hidden units in the single hidden layer, with $h \in \{1, 3, 5, 10, 20\}$.

All input features were standardized within each fold using a `StandardScaler` embedded in a `Pipeline` to prevent data leakage and ensure that no information from the test data influenced the training process. The MSE was used as the loss function.

Outer fold	**ANN**		**Linear regression**		**Baseline**
i	h_i^*	E_i^{test}	λ_i^*	E_i^{test}	E_i^{test}
1	10	6.3218	2154.4347	7.6553	7.6784
2	10	12.0008	2154.4347	11.9514	12.4580
3	10	2.3532	2154.4347	3.4941	3.6367
4	20	15.0367	2154.4347	14.3188	14.6310
5	10	37.6724	2154.4347	40.3020	40.5224
6	10	5.8193	2154.4347	3.4497	3.6214
7	10	4.4311	2154.4347	3.5540	3.6721
8	10	74.3511	464.1589	76.8704	76.5464
9	10	5.0788	464.1589	3.4357	3.2956
10	3	91.5514	464.1589	88.5766	86.6047
Mean outer test errors	**25.4617**		**25.3608**		**25.2667**

Table 3: *Two-level cross-validation table used to compare the three models.*

The results in Table 3 show that all three models achieve comparable performance, with mean test errors of $\overline{E}_{ANN} = 25.46$, $\overline{E}_{Ridge} = 25.36$, and $\overline{E}_{Baseline} = 25.27$.

On average, neither Ridge regression nor the ANN provides a substantial improvement over the baseline model.

2.2 Statistical Evaluation

To assess whether the observed differences were statistically significant, a **paired t-test** was conducted on the ten outer-fold errors. Three pairwise comparisons were evaluated: (1) ANN vs. Ridge, (2) ANN vs. Baseline, and (3) Ridge vs. Baseline. A p-value below 0.05 indicates a statistically significant difference in model performance. The mean difference in error ($\Delta = E_1 - E_2$), the t-statistic, p-value, and the 95 % confidence interval were computed for each pair, as summarized in Table 4.

Model Comparison	***t*-stat**	***p*-value**	**Mean Δ**	**95% CI for Δ**
ANN vs Ridge	0.162	0.8745	0.1009	[–1.1160, 1.3177]
ANN vs Baseline	0.263	0.7985	0.1950	[–1.2583, 1.6482]
Ridge vs Baseline	0.426	0.6801	0.0941	[–0.3388, 0.5270]

Table 4: *Statistical significance of performance differences among the three models.*

The paired t-tests show that none of the model differences are statistically significant ($p > 0.05$ in all cases). Although the Ridge model obtained a slightly lower mean test error, the observed differences fall within the confidence intervals, indicating that these variations are likely due to random fluctuations in the data rather than consistent model advantages. This indicates that both the Ridge and ANN models generalize similarly to the baseline, and that the dataset does not contain clear nonlinear patterns that the ANN can capture more effectively.

Overall, the results indicate that simple linear modeling captures the main structure of the data. The relatively high variance across folds may reflect measurement noise or a limited ability of the selected features to predict CRP values.

3 Classification

3.1 Classification problem & methods

For this part of the report, a classification problem is solved based on our gallstone dataset. The classification problem is binary, where the goal is to predict whether or not a patient has gallstones ($1 = yes, 0 = no$). The prediction is based on a set of clinical and biochemical features described in Project 1 and previously in Table 1.

Three different classifiers are implemented and compared:

- Logistic regression (with λ regularization)
- k-Nearest Neighbors (KNN) (with k regularization)
- Baseline model

To ensure that all features contribute equally to the models, standardization is performed before training. This step is crucial to prevent features with large numerical ranges from dominating.

The baseline model represents the simplest possible classifier. It identifies the most frequent class in the training data and predicts all test samples as belonging to that class. In this project, it serves as a reference point to evaluate the more complex classifiers. The two more complex models are logistic regression and KNN. For logistic regression, model complexity is determined by the regularization parameter λ - smaller values allow the model to fit the data more closely, while larger values increase regularization and limit flexibility. The tested range was $\lambda = [0.001, 0.01, 0.1, 1, 10]$. For KNN, the complexity is determined by the number of neighbors, k - a small k makes the model highly flexible and sensitive to noise, while a larger k results in smoother, more generalized decision boundaries. The tested range was $k = [1, 3, 5, 7, 9]$. Both λ and k were chosen within moderate ranges to balance model flexibility and generalization, avoiding both overfitting and underfitting during the inner cross-validation.

3.2 Two-level Cross-validation

To compare the three models, a two-level cross-validation was performed. The name refers to its twofold process: divided in an **inner cross-validation loop**, which is responsible for selecting the optimal hyperparameters (λ, k), and an **outer cross-validation loop**, which provides an unbiased estimate of each model's generalization error.

Table 5 lists the results of the two-level cross-validation. Logistic Regression achieved the lowest mean test error of 0.238, followed by KNN with mean test error of 0.317, while the Baseline Model performed worst with nearly 50% mean test error (0.496). This indicates that both ML models outperform the Baseline Method, with Logistic Regression providing the best generalization performance overall.

Outer fold	**KNN**		**Logistic regression**		**Baseline**
i	k_i^*	E_i^{test}	λ_i^*	E_i^{test}	E_i^{test}
1	9	0.438	0.001	0.219	0.469
2	9	0.312	1.000	0.156	0.500
3	9	0.344	1.000	0.312	0.500
4	9	0.250	0.001	0.188	0.500
5	9	0.281	1.000	0.312	0.500
6	9	0.312	0.100	0.156	0.500
7	9	0.219	1.000	0.281	0.500
8	5	0.438	1.000	0.344	0.500
9	5	0.281	0.001	0.188	0.500
10	3	0.290	0.001	0.226	0.484
Mean outer test errors	**0.317**		**0.238**		**0.495**

Table 5: *Two-level cross-validation results comparing KNN, Logistic Regression, and the Baseline model on the gallstone dataset. Mean outer test errors for the three classification models.*

Both Logistic Regression and KNN achieved substantially lower classification errors than the baseline, confirming that the selected clinical features carry predictive information about gallstone presence. The baseline model's 50% error corresponds approximately to random guessing in a balanced binary setup.

3.3 Statistical evaluation - McNemar's test

After performing the two-level cross-validation, a pairwise statistical comparison between the classifiers was performed using McNemar's test. This test evaluates whether the observed differences in classification accuracy between two models are statistically significant. The generated p-values and Jeffreys confidence intervals are listed in Table 6.

Contingency tables for each pairwise comparisons were generated to visualize how each classifier agreed or disagreed on the correctly and incorrectly classified samples. From these tables, n_{12} and n_{21} were extracted and are listed in Table 6. n_{12} denotes the number of samples correctly classified by the first model but misclassified by the second, and n_{21} denotes the opposite. The relative difference between these values forms the basis for McNemar's Test.

When evaluating the results in Table 6, it can be concluded that all three pairwise comparisons are statistically significant ($p < 0.05$) with a confidence intervals not containing zero. Both machine learning models outperform the Baseline, with the Logistic Regression also performs significantly better than KNN ($p = 0.0026 < 0.05$). Based

Model Comparison	**n12**	**n21**	$\hat{\theta}$ **(E_theta)**	**95% CI**	**p-value**
Logistic vs KNN	45	20	0.0784	[0.0296, 0.1270]	0.0026
Logistic vs Baseline	112	30	0.2571	[0.1890, 0.3239]	0.0000
KNN vs Baseline	95	38	0.1787	[0.1103, 0.2462]	0.0000

Table 6: *Paired Model comparison results with estimated $\hat{\theta}$, 95% confidence intervals, and significance testing.*

on the McNemar's test, Logistic Regression is therefore recommended for predicting gallstone presence using the given dataset.

3.4 Train logistic regression model

Finally, a logistic regression model was trained to predict the probability of gallstone presence based on the selected features:

$$p(y = 1|x) = \frac{1}{1 + exp - (w^T x + b)}$$

where w are the feature coefficients and b is the bias term.

The optimal regularization parameter found in the previous cross-validation exercise was used, $\lambda = 0.51$. The model was fitted on the standardized data to ensure that all features contributed equally to the weight estimation.

Feature	**Weight**
CRP	2.118442
Vitamin D	-0.735251
TBFR (%)	0.610804
Weight	-0.388079
Height	0.201926
Gender	-0.194298
Age	0.000490

Table 7: *Logistic Regression coefficients (weights) for each feature with regularization parameter $\lambda = 0.51$.*

The result of the trained logisitic regression model is illustrated in Figure 2 and the corresponding weights are listed in Table 7. A positive coefficient indicates that an increase in the corresponding feature raises the probability of gallstones, whereas a negative coefficient decreases it. The relative importance of the feature is reflected by the magnitude of its weight.

From the results, CPR, TBFR (%), and Height (2.12, 0.61, and 0.20, respectively) have positive coefficients, suggesting that higher values increase the likelihood of gallstone formation. CRP shows the highest magnitude, indicating the strongest influence. Age has a positive weight but effectively no influence ($0.000490 = \tilde{0}.0$). Among the negatively weighted features, Vitamin D, Weight, and Gender (-0.74, -0.39, and -0.19) are associated with a reduced probability of gallstones when their values increase.

Overall, both Logistic Regression and KNN demonstrated strong generalization performance, but Logistic Regression achieved significantly better results according to the McNemar test. The model also provides interpretability,

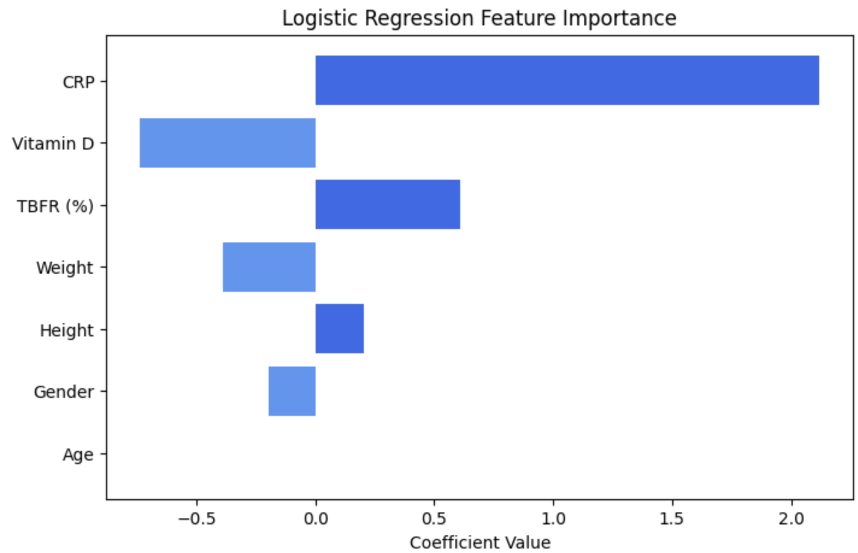


Figure 2: *Weights of features in a trained logistic regression model with* $\lambda = 0.51$.

highlighting the same key features identified in the regression analysis, particularly CRP and TBFR (%), confirming that these physiological markers are consistent indicators of gallstone presence across both tasks.

4 Discussion

The aim of this project was to investigate how biological factors relate to gallstone disease through a classification analysis of gallstone status and regression models of the inflammatory marker CRP. Across all analyses, the focus was on identifying which features had the strongest weights and on evaluating whether more complex machine learning models offered clear benefits over simpler linear approaches.

4.1 Recap and Comparison

In the first regression analysis (Section 1), CRP concentration was modelled using ridge regression with standardized clinical and biochemical predictors. After tuning via cross-validation, the model showed low generalization error and biologically meaningful weights (Table 2). TBFR (%) had the strongest positive association with CRP, while Vitamin D and Height were negative, and Weight, Age, and Gender had minor effects consistent with literature linking adiposity to inflammation and Vitamin D to anti-inflammatory effects. In the model comparison (Section 2), ridge regression, an ANN, and a mean baseline (Table 3) achieved similar test errors, with no significant differences ($p > 0.05$, Table 4). Ridge regression performed slightly better on average, but within confidence intervals, suggesting differences were random. Overall, linear modeling captured the key structure in CRP, while non-linear models offered no added benefit, likely due to noise and limited sample size.

In the classification analysis (Section 3), gallstone presence (yes/no) was predicted from the same predictors using Logistic Regression, KNN, and a baseline classifier within a two-level cross-validation framework (Table 5). Both machine learning models outperformed the baseline, showing that the features hold predictive value. Logistic

Regression achieved the lowest mean test error (0.238) and significantly outperformed both KNN and the baseline ($p < 0.05$, Table 6), indicating that a regularized linear boundary suits this task better than the more noise-sensitive KNN. The final model's weights (Table 7, Figure 2) highlight CRP and TBFR (%) as the strongest positive predictors and Vitamin D as a negative one, while Age had minimal effect. These patterns align with the ridge regression findings, reinforcing that adiposity and inflammation drive gallstone risk, whereas Vitamin D may offer protection.

Taken together, the results across regression and classification point to two main conclusions. First, in this dataset, well-regularized linear models (ridge regression and Logistic Regression) are sufficient to achieve good predictive performance and outperform or match more complex models such as ANN and KNN. Second, the same biological markers in particular CRP and TBFR (%), consistently stand out as the most important predictors in both tasks, reinforcing their relevance as clinical indicators of gallstone-related risk and inflammation.

4.2 Limitations

There are, however, several limitations to consider. The dataset is of moderate size, which increases the uncertainty of parameter estimates and may limit the ability of complex models to generalize. Measurement noise in biochemical variables, unobserved confounders (such as diet, medication use, or genetic factors), and the cross-sectional nature of the data also restrict the strength of causal interpretations. Future work could include external validation on an independent cohort, incorporation of additional risk factors, and exploration of interaction terms within the linear models. Nonetheless, the present analysis indicate that, for the available gallstone data, simple, interpretable linear models combined with carefully chosen features provide a robust and clinically meaningful description of the relationship between metabolic health, inflammation, and gallstone disease.

4.3 Comparison with Existing Literature

A recent study by Esen et al. [3] applied machine learning to predict gallstone disease using biochemical and body composition markers. Similar to our findings, they identified CRP and measures of body fat as central predictors of gallstone presence. They also highlighted Vitamin D as an informative feature in distinguishing between patients with and without gallstones. This is consistent with our results: in both our regression and classification models, Vitamin D showed a negative relationship with CRP and gallstone presence, suggesting a potential protective effect. In Esen et al., Vitamin D is important because it helps the model separate the two groups, while our regression also reveals the direction of the biological relationship. Together, these results support the conclusion that higher adiposity and inflammation increase gallstone risk, while higher Vitamin D levels may have a protective role.

5 Python Code

The Python code used to perform the analyses and generate the results is available on GitHub: `Project 2`.

References

[1] Yuxin Zhang et al. “Inflammation plays a key role in the development of gallstones: a population-based study.” In: *BMC Gastroenterology* 25.1 (2025). Accessed: 23 October 2025, p. 4000. DOI: 10.1186/s12876-025-04000-z. URL: https://bmcgastroenterol.biomedcentral.com/articles/10.1186/s12876-025-04000-z.

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